Foundations: UNIX, Python and PDB Parsing

Instructor guide — preparation, delivery and known issues

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`online/Foundations-unix-python-pdb.ipynb`

At a glance

Runtime	CPU. Do not request a GPU; it buys nothing here and the queue wait is wasted class time.
Wall clock	Two to four minutes, most of it the package install. Budget 75–90 minutes of class time.
Network	Needs RCSB. Pulls `6ane.pdb`, `6ane.cif` and `1y26.pdb` from `files.rcsb.org`.
Prerequisites	None beyond basic Python syntax. This is the entry point to the sequence.
Produces	`data/6ane.pdb`, `data/6ane.cif`, `data/1y26.pdb`, `manifests/lab1_0_lockfile.json`
Quiz items	7, in three blocks

What this notebook is for

This is the on-ramp. Everything downstream — minimization, solvation, dynamics, docking, structure prediction — reads the same PETase crystal structure, so the point of this session is to make students fluent in the file before they ever hand it to a simulation engine.

Three skills carry forward, and they are the ones I actually assess:

1. Reading a PDB file as a *stream of fixed-column records*, not as a magic object.
2. Holding two residue numbering systems in their head at once without mixing them up.
3. Computing a physical property (net charge) from first principles rather than calling a library function and trusting the number.

The enzyme is PETase from *Ideonella sakaiensis*, PDB `6ANE` — a real, 2.02 Å structure of a real enzyme that depolymerises PET plastic. The RNA thread at the end uses the *add* adenine riboswitch, PDB `1Y26`. Both recur in later sessions, so it is worth naming them properly now.

Before class

Open the notebook, set **Runtime → Change runtime type → CPU**, then **Runtime → Run all** and watch it finish. I do this the morning of, because the only thing that reliably breaks is the RCSB download, and if that is down I want to know before thirty students find out.

If RCSB is unreachable, the cells print a notice and continue, but every later cell fails. The fix is to upload the three files by hand into `data/` and re-run from Section 1.

One housekeeping note for the students: Colab runs on a disposable virtual machine. If the session disconnects, `data/` is gone and they must start from the top. Say this out loud before they start, not after someone loses an hour.

Section-by-section

Section 0 — environment

A single `pip install` cell pinning Biopython 1.83, jupyterquiz 2.8.1, py3Dmol 2.1.0 and gemmi 0.6.5, plus a `COURSE_SEED = 42`. Takes about 40 seconds. Nothing to teach here, but do point out that the pins are deliberate: an unpinned notebook is an unreproducible notebook.

Section 1 — the shell, and why columns matter

Downloads `6ANE` and then interrogates it with `wc`, `grep` and `awk`. The numbers students should see:

```
6869   lines in the file
5808   ATOM records
 475   HETATM records
 789   ATOM records matching " CA "
 474   HOH waters
```

Then a gemmi pass over the mmCIF version pulls the refinement metadata: resolution 2.02 Å, R-work 0.177, R-free 0.217, space group C 2 2 2_1.

Two things are worth stopping on. First, `grep " CA "` works only because the PDB format is column-fixed, and it would break on a calcium ion (`CA` as an element). Ask them why the spaces are inside the quotes. Second — and this is the good one — **789 does not equal the 787 residues the next section reports**. The two extra alpha-carbons are the alternate conformers in chains B and C. I let them find that discrepancy themselves in Section 2 rather than pre-empting it here.

The `awk` cell slices columns by field number. On `6ANE` it behaves, but tell them that field-splitting on whitespace is fragile for PDB files in general: when a coordinate column runs together with its neighbour, `$5` silently becomes the wrong thing. That is the argument for the column-slicing parser they write in Section 2.

Section 2 — a parser, written by hand

An `Atom` class slicing by character position, then a nested dictionary of chain → residue → atoms. Expected output:

```
Parsed 6283 total atoms from 6ANE.
Chain A: 262 amino acid residues resolved.
Chain B: 264 amino acid residues resolved.
Chain C: 261 amino acid residues resolved.
Total residues exhibiting conformational heterogeneity (altLocs): 2
  Chain B, Residue #158 has alternate conformers: ['A', 'B']
  Chain C, Residue #158 has alternate conformers: ['A', 'B']
```

Residue 158 is the tryptophan. It is the only residue in the whole structure modelled in two conformations, and it is the catalytically interesting one — the aromatic clamp that stacks against the terephthalate ring of the substrate. The crystallographers saw it in two positions because it genuinely moves. That is not a coincidence to skip past; it is the single best example in this course of experimental data carrying dynamic information.

Section 3 — net charge and the titration curve

Henderson–Hasselbalch over a standard pKa table, applied residue by residue to chain A.

```
PETase Chain A sequence length: 262 amino acids
Calculated Net Charge at physiological pH 7.40: +6.288 e
Calculated Isoelectric Point (pI): 9.72
```

A positive number surprises people — "it's a hydrolase, shouldn't it be acidic?" — so have the answer ready: chain A carries 13 Arg and 7 Lys against 9 Asp and 4 Glu. Count them on the board if you have time; it takes a minute and it makes the number concrete.

Histidine is the teaching point. At pH 7.4 with pKa 6.0 it is about 4% protonated, so it contributes almost nothing to the net charge, yet it is the catalytic base. Charge and chemistry are not the same question.

> **Carry this number forward.** `+6.29 e` is computed here from the sequence. The solvation notebook later in the sequence hard-codes `net_charge=-2` for the same chain and adds counter-ions on that basis. They cannot both be right, and this one is right. I flag it in that guide too; if you teach the two sessions back to back, make it an exercise rather than an erratum.

Section 4 — rendering, and the numbering contract

py3Dmol renders chain A with the catalytic machinery highlighted, followed by a table mapping the two numbering systems. This is the part students get wrong for the rest of the semester if it is not nailed down here.

ROLE	UNIPROT A0A0K8P6T7	6ANE COORDINATES
Nucleophile (Ser)	160	**133**
Electrostatic anchor (Asp)	206	**179**
General base (His)	237	**210**
Aromatic clamp (Trp)	185	**158**

The offset is 27, the length of the cleaved signal peptide. I have verified every row of this against the deposited coordinates: in `6ANE`, residue A133 is SER, A179 is ASP, A210 is HIS and A158 is TRP. Residue A160, by contrast, is a different serine entirely, and A206 and A237 are both asparagine. If a student selects residue 160 expecting the nucleophile, they get a plausible-looking serine in the wrong place and no error message. Watch for it.

The notebook calls the 160/206/237 column "mature sequence numbering". That label is backwards — 160 is the *precursor* (UniProt) number, and the deposited 133 is the mature number, counting from the first residue after the signal peptide. The arithmetic in the cell is correct; only the word is wrong. The databases session states it the right way round ("UniProt #160, deposited PDB #133"), so if you teach them in order, correct it here verbally and the next session will reinforce the fix.

Chain A is numbered 2–263 in the coordinate file, not 1–262. Mention it. It is the root cause of an off-by-two indexing error in the dynamics notebook later on.

Section 5 — the RNA thread

Downloads `1Y26` and tabulates the adenine binding pocket. The ligand is `ADE 90` in chain X, and the three bases listed in the table are genuine close contacts — I measured them: U74 at 2.65 Å, U51 at 2.77 Å, U22 at 2.78 Å, closest heavy atom to ligand in each case. U74 is the Watson–Crick discriminator that reads adenine over guanine, which is the detail the docking session's report question comes back to.

Section 6 — report questions

Two: a histidine-pKa sweep, and an altLoc hunt across chains B and C. Both are well posed and both are answerable with code already in the notebook.

For the altLoc question, the answer is that the only alternate conformers in `6ANE` are at residue 158 in chains B and C — so the honest finding is "almost none, and the one that exists is functional, not surface noise". A student who reports that has done the work. A student who reports a long list has a bug in their altLoc test, most likely treating the blank altLoc field as a conformer label.

Quiz answer key

#	TOPIC	CORRECT ANSWER
1	ATOM vs HETATM	ATOM = standard polymer residues; HETATM = ligands, ions, waters
2	`grep " CA " | wc -l`	Count of residues with a resolved alpha-carbon, across all chains
3	Occupancy 0.60, altLoc A	That atom sits in conformation A in ~60% of unit cells
4	His at pH 7.4	pH is 1.4 units above its pKa, so it is ~96% neutral imidazole
5	Definition of pI	The pH at which summed positive and negative charges cancel
6	Trp185 "wobbling" clamp	π-stacks with terephthalate; modelled in alternate conformers
7	RNA polyanion	Each phosphodiester has an ionisable OH with pKa < 2, so −1 e per nucleotide

Item 2 is the one I would quibble with as written. The literal output is 789, while the structure holds 787 residues, because two residues contribute two CA records each. The answer is right in spirit and off by two in fact. I treat the discrepancy as a discussion prompt rather than fixing the question, but if you are grading strictly, say so in advance.

Known issues in this version

- **"Mature" and "precursor" are swapped** in the Section 4 numbering note, as described above. Arithmetic fine, wording wrong.
- **789 vs 787 alpha-carbons.** Unremarked in the notebook. Good discussion, bad surprise.
- **The net charge disagrees with the solvation notebook** (+6.29 e here, `-2` hard-coded there). This one is correct.
- **No primary citation for `6ANE`.** It is Fecker et al., *Biophysical Journal* 114:1302–1312 (2018), `10.1016/j.bpj.2018.02.005`. I give it to students when I introduce the structure.
- The py3Dmol viewers render nothing if the notebook is opened in anything other than Colab or classic Jupyter. Not a problem in Colab; worth knowing if a student emails you a screenshot of an empty box from VS Code.

Troubleshooting

SYMPTOM	CAUSE	FIX
`FileNotFoundError: data/6ane.pdb`	Download cell skipped, or session restarted	Re-run from Section 1
Empty 3D viewer	Not running in Colab, or output cleared	Re-run the render cell; use Colab
`ModuleNotFoundError: gemmi`	Section 0 not run	Run Section 0, then re-run the failing cell
Residue 160 is the wrong serine	Numbering mix-up	Coordinates use 133/179/210/158
Everything gone after lunch	Colab VM recycled	Re-run all; it takes ~4 minutes